Md. Nazim Hossain

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Summary

Passionate Computer Science student with hands-on experience in full-stack MERN development, Artificial Intelligence, and Natural Language Processing. Adaptive to applying machine learning algorithms, transformer models, and software engineering principles to solve complex real-world problems. Looking for opportunities to work at the architectural level and build useful tools.

Education

2022 - 2026 **BSc in Computer Science and Engineering (CSE)** (Overall CGPA: 3.78/4.00)
Brac University, Merul Badda, Dhaka CSE Major CGPA: 3.90/4.00

2019 - 2021 **Higher Secondary Certificate (HSC)** (GPA: 5.00/5.00)
Birshreshtha Munshi Abdur Rouf Public College, Peelkhana, Dhaka

2017 - 2019 **Secondary School Certificate (SSC)** (GPA: 5.00/5.00)
Willes Little Flower School And College, Kakrail, Dhaka

Skills

AI & Machine Learning: NLP, Transformer Models (BERT, LLMs), Domain-Adaptive Pretraining (DAPT), Explainable AI (LIME, SHAP), CNNs, Transfer Learning
Languages: Python, Java, C++, C, JavaScript, SQL, PHP
Web Technologies: MERN Stack (MongoDB, Express.js, React, Node.js), React Native, HTML, CSS
Tools & Concepts: Git, GitHub, MVC Architecture, OOP, Data Structures, Algorithms, Linux

Projects & Research

NLP-Based Mental Health Detection (Thesis) *Brac University* – [Live Demo]

Engineered a low-resource Bangla NLP framework achieving an **87.20% weighted F1** on depression severity classification. Designed a **decoupled LLM synthetic augmentation scheme** (Qwen3-32B) during DAPT/TAPT to prevent label noise, validated models via LIME/SHAP explainability, and built a SigLIP cross-attention multimodal extension.

GardeniAR [Github]

Developed a full-stack MERN application featuring a dual-mode AR Weed Identification tool using AI integration (Gemini Multimodal) and React-Webcam for real-time invasive species detection. **2nd Runners Up** - CSE471 Project Demonstration, Brac University.

Meal Sync [Github]

Built a dynamic meal subscription platform connecting consumers with home chefs. Implemented using the MERN Stack and MVC Architecture to bring convenience to both businesses and clients.

Song-Lyrics Genre Classification [Github] — [Live Demo]

Built a 3-model soft-voting transformer ensemble (BERT, RoBERTa) in PyTorch and Hugging Face to classify 53K+ lyrics across 6 genres. Implemented class-weighted cross-entropy loss to address severe class imbalance, achieving 0.56 Macro F1 at 0.66 accuracy and deploying a live demo on Hugging Face Spaces.

hyprGesture (Embedded Linux Gesture Controller) [Github]

Developed a touchless Linux desktop controller interfacing an Arduino UNO and PAJ7620 sensor with the Hyprland window manager. Built a Python serial listener with non-blocking cooldown timing and asynchronous process management (`subprocess`) to execute 27 gesture commands across 3 operational modes.